

week-2(Pravalika 2520030402)

Develop a C program that uses open(), read(), write() and close() to copy the contents of one file to another. Explain the control transition between user space and kernel space. Use strace to trace the system calls generated by cat sample.txt

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source, destination;
    char buffer[100];
    int bytes;

    source = open("sample.txt", O_RDONLY);

    if (source < 0)
    {
        printf("Error opening source file\n");
        return 1;
    }

    destination = open("copy.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0)
    {
        printf("Error creating destination file\n");
        close(source);
        return 1;
    }

    while ((bytes = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytes);
    }

    close(source);
    close(destination);

    printf("File copied successfully\n");
}
```

Output:

```
rikki@BOOK-GJU4MP9458:~$ echo "This is a sample file for Practical 2." > sample.txt
rikki@BOOK-GJU4MP9458:~$ gcc practical2.c -o practical2
rikki@BOOK-GJU4MP9458:~$ ./practical2
File copied successfully
rikki@BOOK-GJU4MP9458:~$ cat copy.txt
This is a sample file for Practical 2.
rikki@BOOK-GJU4MP9458:~$
```